Junhong Tong
Curriculum Vitae
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EDUCATION

Master of Computer Science, Queens College, CUNY, USA **2026**

Bachelor of Electronic Engineering, South China Agricultural University, China **09/2019 – 06/2023**

Experience

Research Assistant, University of North Carolina at Chapel Hill **Present**
ACM Lab (acmlab.unc.edu)

Research Interest

Healthcare, Machine Learning, Neuroimaging, Coding Theory, Natural Language Processing, LLM.

Publications

- [1] Yuchun Zou, Junhong Tong, Jun Li. Text-Preserving Lossy Text Compression: A Study of Strategic Deletion and LLM Reconstruction. AACL-IJCNLP 2026
- [2] Yiyao Chen, Yucheng Li, Junhong Tong, Shaoqi Wang, Kunhao Zhou, Ziquan Wei, Monica Murea, Marissa DiPiero, Tingting Dan, Guorong Wu. NeuroPilot: An Agent-Driven Smart Pipeline for Processing, Quality Control, and Managing Neuroimages. NeurIPS 2026 Workshop on World Models for High-Stakes Health (WMHS 2026)
- [3] Ran Ren, Junhong Tong, Yunxi Kong, Yiyao Chen, Yucheng Li, Kunhao Zhou, Shaoqi Wang, Yuxiang Tao, Shuheng Cao, Zhihao Fan, Marissa DiPiero, Tingting Dan, Guorong Wu. CyberNeuro: A Privacy-Preserving Agentic Workbench for Cohort-Scale Neuroimage and Clinical Data Analysis. arXiv preprint arXiv:2607.28841, 2026
- [4] Junhong Tong, Yuchun Zou, Jun Li. Can Language Models Help Error-Correcting Codes for Text, and When? (Under review at ARR October 2026)

Research Experience

CyberNeuro: AI Agent Platform for Neuroimaging Research (UNC ACMLab) Supervised by Guorong Wu

- Built an AI agent platform whose assistant, Wanda, lets researchers run neuroimaging analyses in natural language, without installing complex toolchains or writing cluster job scripts.
  Demo: wanda-landing-page-orcin.vercel.app
- Designed the workflow: the agent maps each request to one of 58 Skills (an instruction document plus an Apptainer SIF image) covering structural preprocessing, DICOM-to-BIDS, structural MRI QC, white-matter normative scoring, DTI/tractography, functional connectivity, and statistics and paper figures, then generates and submits Slurm jobs to the cluster's CPU/GPU nodes.
- Developed a SolidJS web app with a chat interface, file browser, and visualization panel, with Skills shown as pills in the input box; outputs are routed by file type (NIfTI to a slice viewer, FreeSurfer surfaces to PyVista 3D, fiber tracts to FURY/TrackVis, connectivity matrices to network graphs).
- Delivered a 5-step structural MRI QC workflow (raw image, skull stripping, registration, tissue segmentation, cortical surface) with per-subject pass/fail review pages and in-browser brain-mask editing; exported decisions drive the agent's repair or rerun.
- Co-designed a multi-agent framework (Planner, Validator, Dispatcher, Reporter); with a locally hosted 7B model, raised held-out tool-use accuracy from 40.3% to 69.4% and completed a 10-cohort workflow suite with $\sim$10.6% of the per-volume tokens of the NeuroClaw baseline.

Text-Preserving Lossy Text Compression: A Study of Strategic Deletion and LLM Reconstruction Supervised by Jun Li

- Proposed a lossy semantic text compression framework combining token deletion and LLM-based reconstruction.
- Designed multiple deletion strategies (WordFreq, entropy-based, and hybrid methods) to optimize the compression–quality trade-off.
- Fine-tuned Llama-3.2-3B-Instruct using QLoRA for reconstruction, achieving competitive performance with Gemini 2.0 Flash.

- Evaluated reconstruction quality using BERTScore, CER, and ROUGE-L across multiple datasets, including BBC News and Chinese corpora.
- Conducted large-scale experiments under varying compression rates (10%–90%) to analyze robustness and generalization.

Project

Design and Implementation of an Automobile Anti-collision Warning System **March 2023 – Jun 2023**

- Engineered an end-to-end forward collision warning system, from schematic design and hardware implementation to firmware development on an 8-bit MCU.
- Implemented a sensor fusion algorithm to process and combine data from ultrasonic and infrared sensors, significantly enhancing object detection accuracy and minimizing false positives.
- Programmed a multi-stage warning logic with tiered distance thresholds, triggering escalating audible and visual alerts to clearly signal imminent collision risks.
- Established a wireless communication link using the UART protocol to broadcast real-time distance data to a remote monitoring unit, enabling advanced remote alerts.

Cloud Computing **March 2025 – April 2025**

- Built a multi-AZ AWS (EC2) infrastructure and deployed containerized applications on Kubernetes with zero-downtime rolling updates.
- Ran distributed analytics on Hadoop (HDFS) & Spark, and benchmarked a 6-node sharded Redis Cluster's latency and throughput under automated failover.

Technical Skills

Languages & Frameworks: Python, TypeScript, C, Bash, Bun, SolidJS, Hono
ML & LLM: LLM Agents, QLoRA Fine-tuning, YOLOv8, LightGBM
Infrastructure: Docker, Kubernetes, Slurm, Apptainer, AWS, Hadoop, Spark, Redis, GitHub Actions
Neuroimaging: FreeSurfer, fMRIPrep, XCP-D, BIDS

Award

- Received the "Excellent Volunteer" Award for critical operational support and team coordination during a national-level cheerleading competition. **12/2019**
- Earned the "Outstanding Officer" Award in the Student Association for exemplary leadership in organizing major campus-wide events. **06/2020**
- Awarded the Outstanding Student Leader Scholarship, a highly selective, merit-based honor recognizing exceptional leadership and academic excellence. **09/2021**